



END-USER TRAINING GUIDE

MariSoft-AIS Structures

Step-by-step operator training with guided practice and competency sign-off

LEVEL	FORMAT	DURATION	VERSION
New end user	Instructor-led / self-study	4-6 hours	1.0

Application: MariSoft-AIS Structures v11.0.0

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1. Training overview

This guide teaches a new end user to complete a controlled inspection session in MariSoft-AIS Structures. It focuses on actions the trainee must perform, the result to expect, and the checks required before moving forward.

1.1 Learning outcomes

- Connect to the correct database, asset, schedule, and workpack.
- Navigate the component hierarchy and confirm the selected context.
- Start, complete, and end a task while recording structured data and evidence.
- Create and review events, convert an event to an anomaly, and record follow-up actions.
- Use video, Remote DVR, point-cloud, PDF, and 3D tools safely.
- Close a session with verification, backup/export, and handover checks.

1.2 Training environment

Checkpoint: Confirm each item before continuing:

- [] A training or approved non-production database is available.
- [] The trainee knows the assigned Region, Site, Asset, Schedule, and Workpack.
- [] Sample video, images, PDF drawing, point-cloud model, and 3D drawing are accessible.
- [] The trainer has confirmed which administrator-only commands may be demonstrated.

Safety rule: Do not practice Delete All, database creation, workpack transfer, renumbering, or external-ID updates in a live production database.

2. Module 1 - Launch and orientation

Objective: identify the application areas and confirm the version before connecting to data.

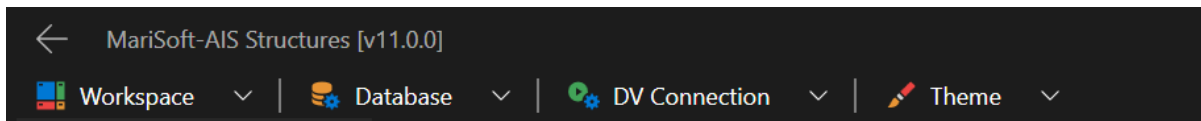


Figure 1. Application banner and global menus

1 Start the application Open MariSoft-AIS Structures from the approved shortcut.

Expected result: The title bar displays MariSoft-AIS Structures and the expected application version.

2 Identify the global menus Locate Workspace, Database, DV Connection, and Theme across the top banner.

3 Identify the navigation panel Use the left panel to locate Database Connection, Inspection, and Templates.

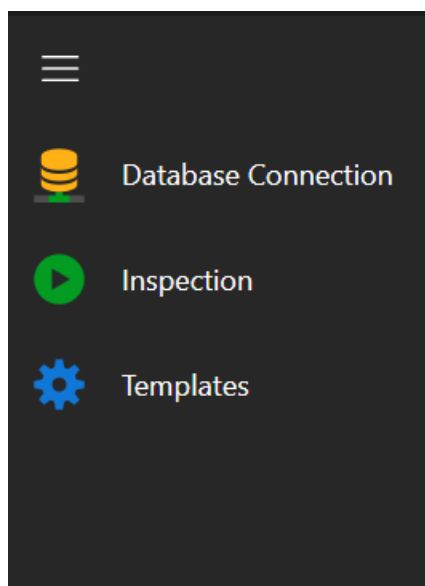


Figure 2. Primary navigation panel

4 Understand command state Notice that Inspection remains unavailable until valid data is loaded.



Figure 3. Status bar context

5 Locate the status bar Find Server, Database, Asset, Schedule, Workpack, and NodeJS Running at the bottom.

Expected result: You can point to every context field without opening a menu.

PRACTICE: Interface orientation

Your task: Name the four global menus, three navigation destinations, and six status indicators.

Successful when: All areas are identified correctly without assistance.

3. Module 2 - Connect to the correct data

Objective: load an approved database and verify the complete inspection context.

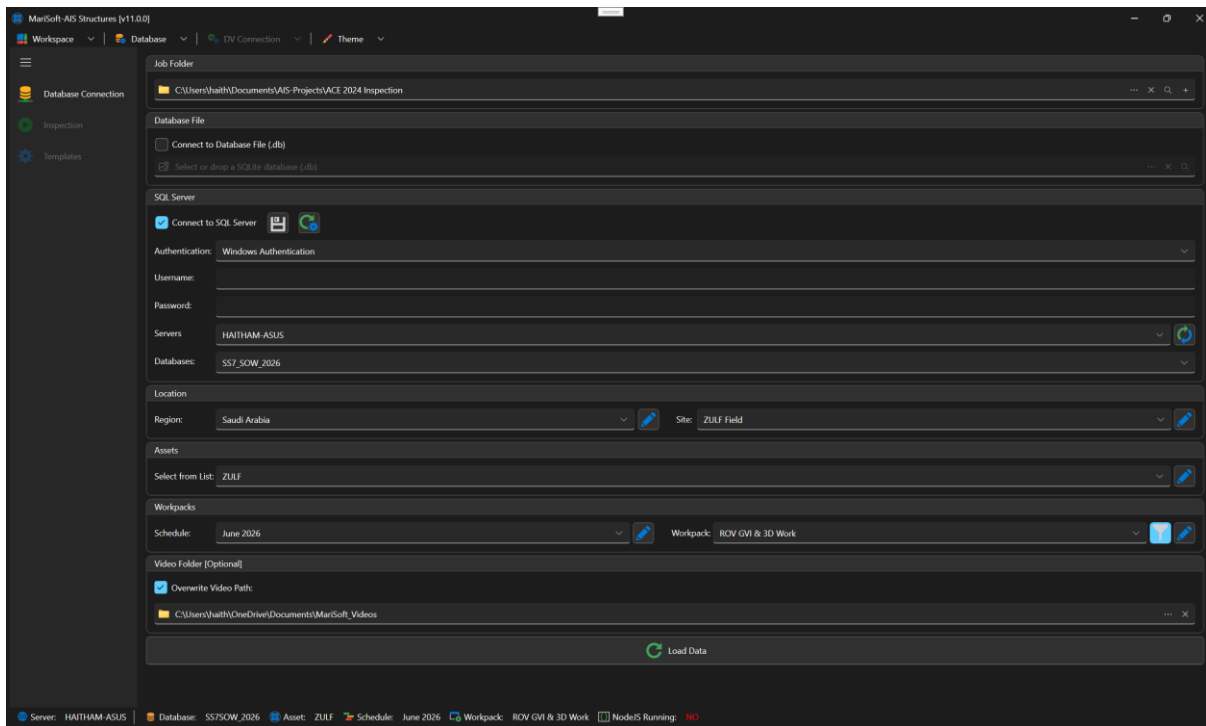


Figure 4. Database Connection screen

- 1 Select the Job Folder** Browse to or drop the approved job folder. This is the working location for the inspection project.
Expected result: The full approved folder path is displayed.
- 2 Choose one database method** For SQLite, enable Connect to Database File and select the .db file. For SQL Server, enable Connect to SQL Server.
- 3 Set authentication** For SQL Server, choose Windows Authentication when approved, or enter the provided username and password. Never include passwords in screenshots or support messages.
- 4 Select server and database** Refresh server availability if required, then choose the authorized server and database.
Expected result: The intended server and database names are visible.
- 5 Select location** Choose Region and Site in that order.

6 Select inspection scope Choose Asset, Schedule, and Workpack in that order. Use the workpack filter only when the list is long.

7 Set the optional video folder Enable Overwrite Video Path only when the approved video location differs from stored paths; browse to the correct folder.

8 Load Data Choose Load Data and wait until the Inspection destination becomes available.

Expected result: The status bar shows the chosen server/database, asset, schedule, and workpack.

Checkpoint: Confirm each item before continuing:

- [] The database name is correct.
- [] The asset and workpack match the training assignment.
- [] The video folder is reachable when video is required.
- [] No connection error remains on screen.

PRACTICE: Controlled connection

Your task: Connect to the trainer-assigned database and workpack, then read the five main context values from the status bar.

Successful when: Every value matches the assignment sheet and Inspection opens successfully.

4. Module 3 - Navigate components and workpacks

Objective: select the correct component and interpret the workspace before creating records.

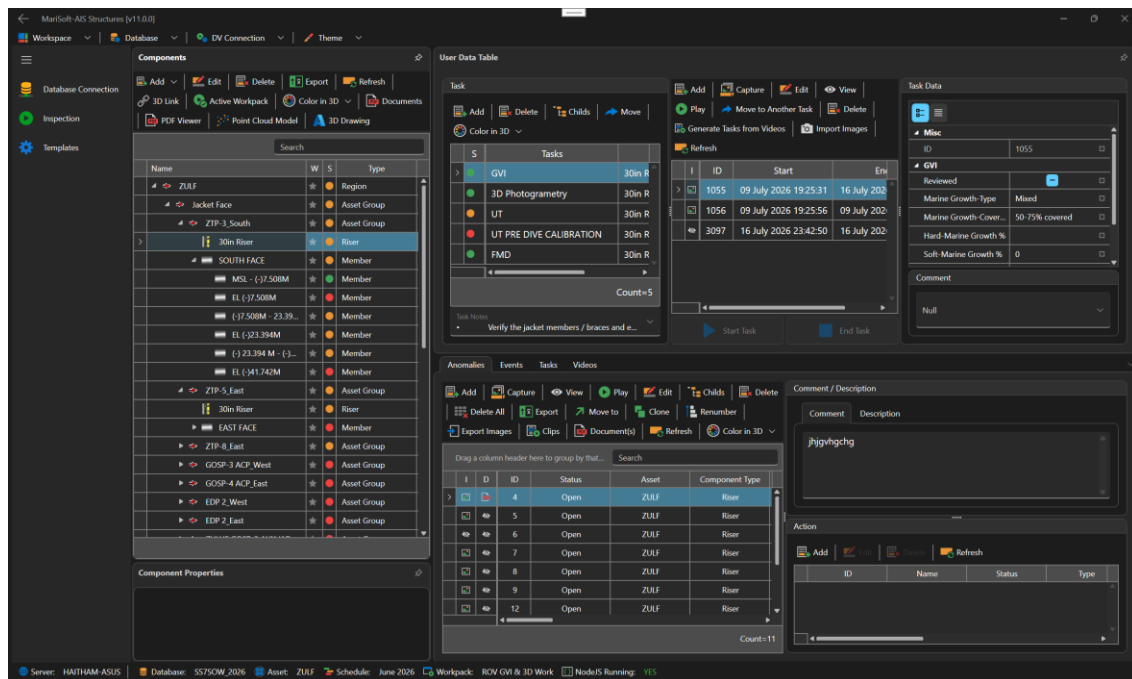


Figure 5. Inspection workspace

- 1 Open Inspection** Choose Inspection from the navigation panel after data has loaded.
- 2 Expand the component hierarchy** Use the arrows beside Region, Asset Group, Site, Discipline, and equipment nodes until the assigned component is visible.
- 3 Select the component** Choose the exact equipment/member node required by the training scenario.

Expected result: The selected row is highlighted and task information updates.

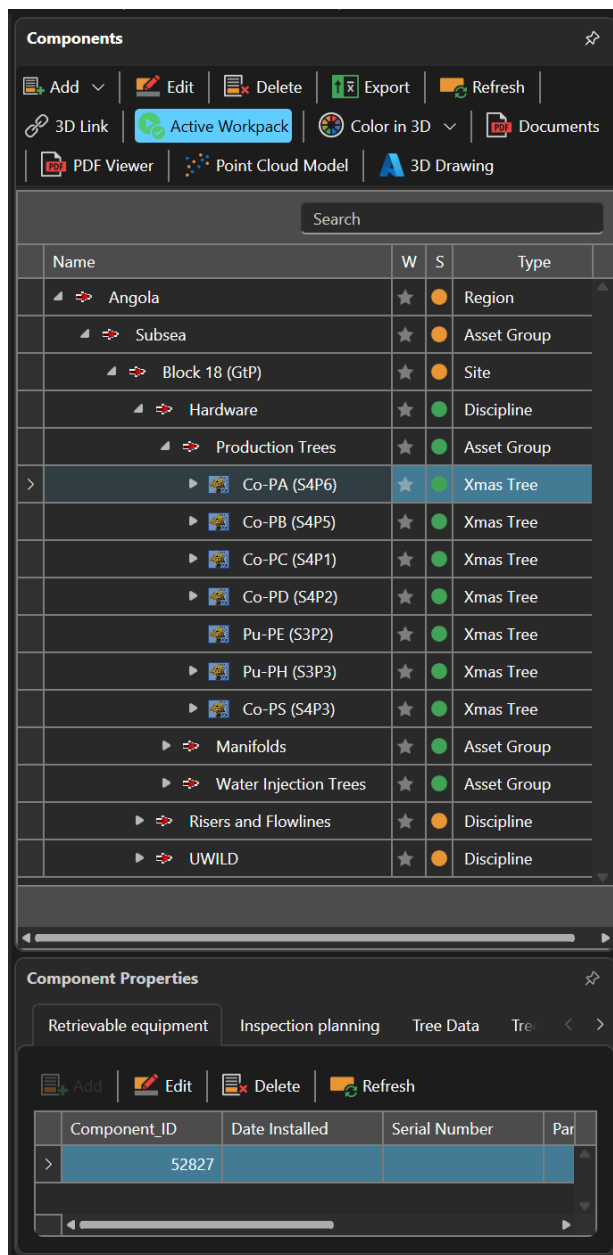


Figure 6. Component hierarchy and properties

- 4 Review component type and status** Check the Name, Type, status indicator, and relevant Component Properties tabs.

5 Use Search when required Enter part of a component name, confirm the result, then clear the search before browsing elsewhere.

6 Confirm Active Workpack Verify the active-workpack control and bottom status bar before recording work.

PRACTICE: Component selection

Your task: Find a component specified by the trainer, identify its parent hierarchy, component type, and available task types.

Successful when: The correct component remains selected and the trainee can explain the hierarchy path.

5. Module 4 - Perform a task

Objective: create or select a task record, start it, complete the required data, attach evidence, and end it correctly.

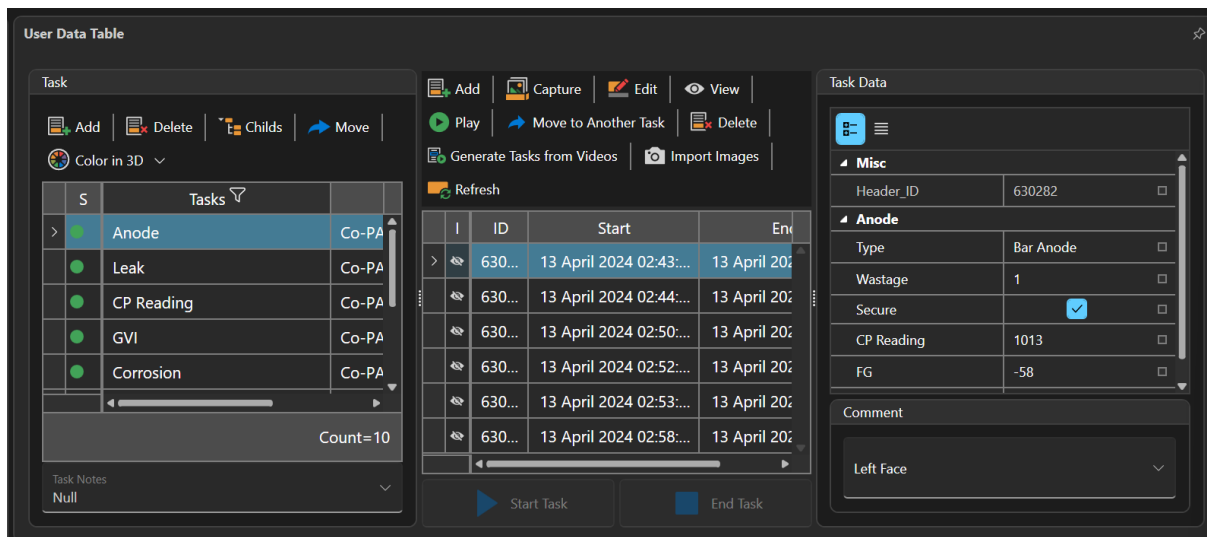


Figure 7. Task selection, records, and Task Data

1 Select the task type In the Task list, choose the required task such as GVI, CP Reading, Anode, or Corrosion.

2 Review Task Notes Read the task instruction before creating or starting a record.

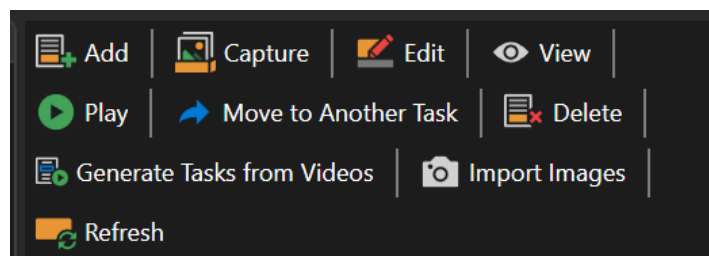


Figure 8. Task-record commands

3 Create or select the task record Choose Add for a new record, or select the existing record assigned by the trainer.

Expected result: The intended record is highlighted and its ID/data are visible.

4 Start Task Confirm the component, task type, and video position, then choose Start Task.

Expected result: A valid Start time appears and the task is shown as active.

5 Complete Task Data Enter each required field in the Task Data panel. Use approved units, codes, check boxes, and terminology.

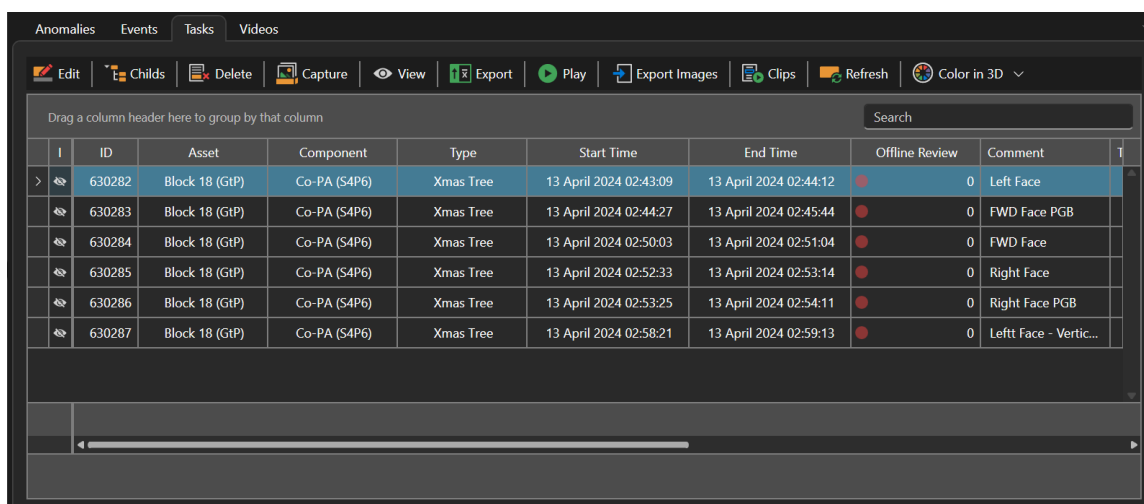
6 Add a factual comment Record the location/orientation and observation. Avoid unsupported conclusions or ambiguous abbreviations.

7 Capture or import evidence Use Capture for the current media frame or Import Images for an approved file. Confirm the evidence belongs to this task.

8 Review media Use View or Play to confirm image/video quality and timing. Use Move to Another Task only if the record was created under the wrong task.

9 End Task Review the data and evidence, then choose End Task.

Expected result: End time is later than Start time and the record remains in the Tasks tab after Refresh.



	I	ID	Asset	Component	Type	Start Time	End Time	Offline Review	Comment
>		630282	Block 18 (GtP)	Co-PA (S4P6)	Xmas Tree	13 April 2024 02:43:09	13 April 2024 02:44:12	0	Left Face
		630283	Block 18 (GtP)	Co-PA (S4P6)	Xmas Tree	13 April 2024 02:44:27	13 April 2024 02:45:44	0	FWD Face PGB
		630284	Block 18 (GtP)	Co-PA (S4P6)	Xmas Tree	13 April 2024 02:50:03	13 April 2024 02:51:04	0	FWD Face
		630285	Block 18 (GtP)	Co-PA (S4P6)	Xmas Tree	13 April 2024 02:52:33	13 April 2024 02:53:14	0	Right Face
		630286	Block 18 (GtP)	Co-PA (S4P6)	Xmas Tree	13 April 2024 02:53:25	13 April 2024 02:54:11	0	Right Face PGB
		630287	Block 18 (GtP)	Co-PA (S4P6)	Xmas Tree	13 April 2024 02:58:21	13 April 2024 02:59:13	0	Left Face - Vertic...

Figure 9. Completed task records

Checkpoint: Confirm each item before continuing:

- [] Correct component and task type.
- [] Valid Start and End times.
- [] All required structured fields completed.
- [] Comment is factual and location-specific.
- [] Evidence opens and matches the task.

PRACTICE: Complete one inspection task

Your task: Create a task record, start it, enter the trainer-provided reading and comment, attach one image, end it, and refresh.

Successful when: The record is complete, correctly timed, and visible under the intended task and component.

6. Module 5 - Record and manage events

Objective: record a coded inspection event, attach evidence, and convert it to an anomaly when required.

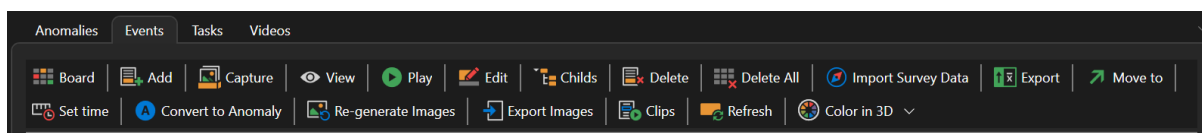


Figure 10. Events command bar

- 1 Open Events** Choose the Events tab beneath the inspection workspace.
- 2 Create the event** Choose Add, then select the correct component, Primary Event/Code, and Secondary Event/Code.
- 3 Set the event time** Use Set time when the current video position represents the observation.
- 4 Attach evidence** Use Capture, Export Images, or Clips as required. Use Play and View to confirm the result.
- 5 Save and refresh** Save the event and choose Refresh.

Expected result: The event has a unique ID and appears under the correct component.

	I	ID	Asset	Component Type	Component	Primary Event	Primary Code	Secondary Event	Secondary Code
>		47526	Block 18 (GtP)	Xmas Tree	Co-PA (S4P6)	Coating damage	CD	Coating damage	CD
		47527	Block 18 (GtP)	Xmas Tree	Co-PA (S4P6)	Corrosion	CR	Corrosion	CR
		47528	Block 18 (GtP)	Xmas Tree	Co-PA (S4P6)	Seeps, weeps and...	LE	Seeps, weeps and...	LE
		47529	Block 18 (GtP)	Xmas Tree	Co-PA (S4P6)	Corrosion	CR	Corrosion	CR
		47530	Block 18 (GtP)	Xmas Tree	Co-PA (S4P6)	Anode wastage	AW	Anode wastage	AW

Count=5

Figure 11. Event records and coding

6 Convert when required Select the event and choose Convert to Anomaly. Confirm the destination component and review copied description/evidence.

7 Complete the anomaly Enter risk, likelihood, status, and any required anomaly-specific fields; do not assume these were inferred from the event.

PRACTICE: Event-to-anomaly workflow

Your task: Create a coating-damage event with evidence, verify its code, then convert it to an anomaly in the training database.

Successful when: Both records are traceable, correctly linked, and use approved codes.

7. Module 6 - Manage anomalies and actions

Objective: create a complete anomaly record, manage reportable images, and add a follow-up action.

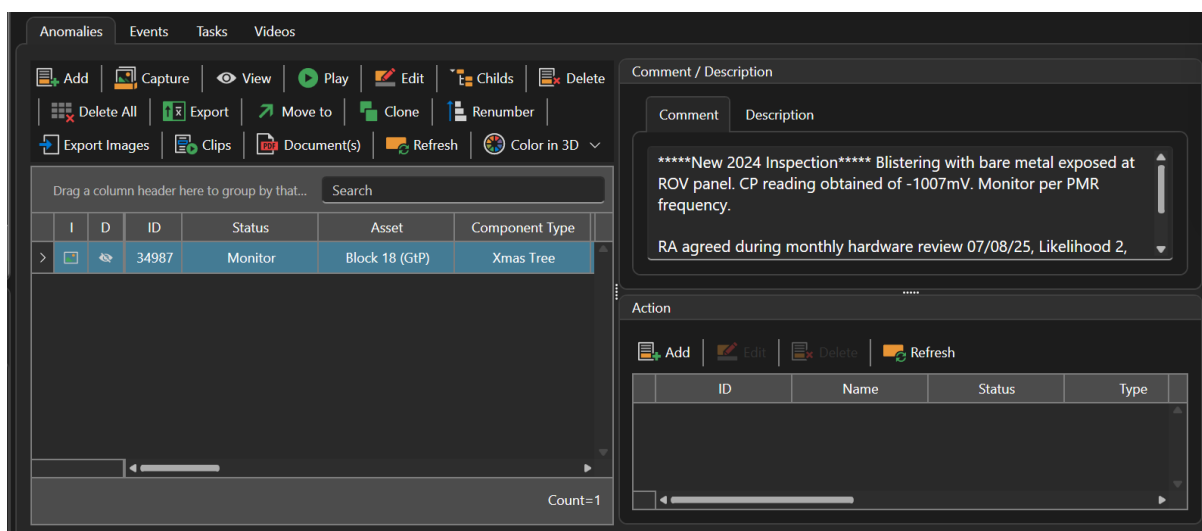


Figure 12. Anomaly record, narrative, and actions

1 Select or add the anomaly Use Add for a new record or select the converted anomaly from Module 5.

2 Complete the classification Verify anomaly code/category, asset, component type, component, workpack, and year/reference fields.

3 Describe the observation Write a factual description including location, extent, measurement, and evidence reference where available.

4 Set risk and status Choose the approved likelihood, consequence/risk, and workflow status. Do not close a record without the required review.

5 Attach and review evidence Use Capture, Documents, Clips, or Export Images as required.

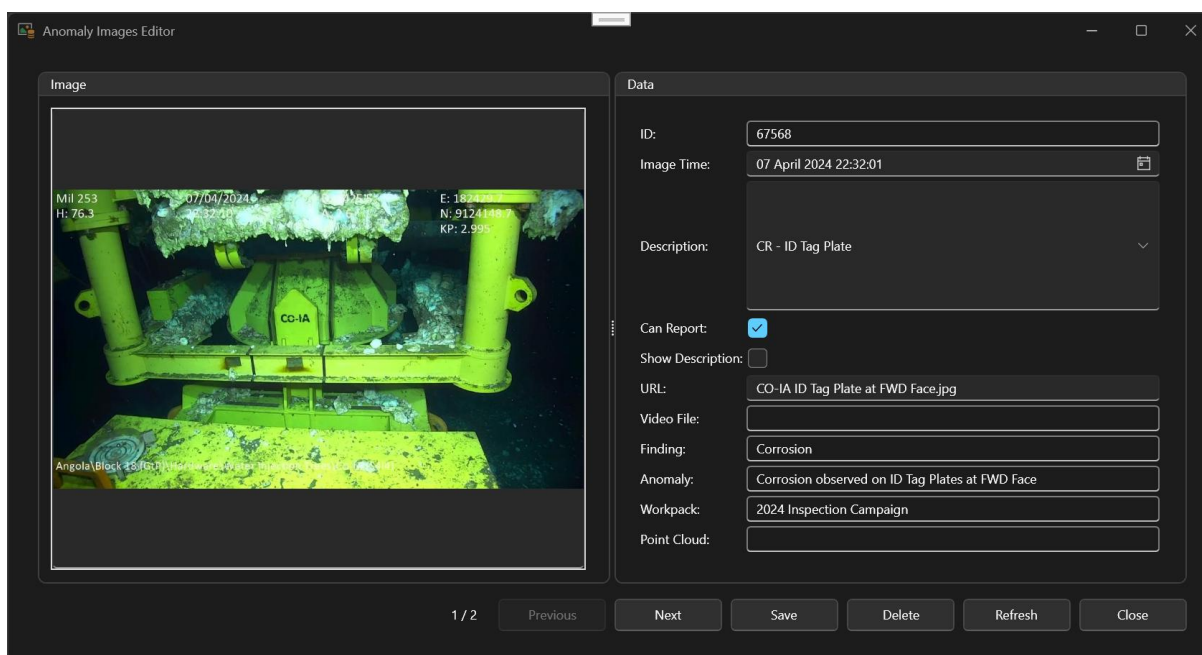


Figure 13. Anomaly Images Editor

- 6 Review anomaly image metadata** Check Image Time, Description, URL, Finding, Anomaly, Workpack, and Point Cloud relationship.
- 7 Set reporting controls** Enable Can Report only for approved evidence. Enable Show Description only when the caption should appear in output, then Save.
- 8 Add a follow-up action** In the Action panel choose Add, enter the action name/type, status, owner or responsible role, and due/completion dates as applicable.
- 9 Verify the record** Refresh and confirm the anomaly, image, and action remain linked to the correct component/workpack.

Checkpoint: Confirm each item before continuing:

- [] Anomaly code, risk, and status follow the approved process.
- [] Description is measurable and evidence-based.
- [] Reportable image settings are deliberate.
- [] Action has a clear requirement and status.

8. Module 7 - Video and Remote DVR

Objective: locate or import video records, play the correct media, and operate the approved Remote DVR connection.

ID	ROV	File Name	File Name 2	File Name 3	Start Time	End Time	Duration
2704		2019-04-12 0451...			12 April 2019 04:5...	12 April 2019 04:5...	0
2703		2019-04-12 0449...			12 April 2019 04:4...	12 April 2019 04:5...	0
2702		2019-04-12 0447...			12 April 2019 04:4...	12 April 2019 04:4...	0
2701		2019-04-12 0445...			12 April 2019 04:4...	12 April 2019 04:4...	0
2700		2019-04-12 0444...			12 April 2019 04:4...	12 April 2019 04:4...	0
2699		2019-04-12 0442...			12 April 2019 04:4...	12 April 2019 04:4...	0
2698		2019-04-12 0440...			12 April 2019 04:4...	12 April 2019 04:4...	0
2697		2019-04-12 0438...			12 April 2019 04:3...	12 April 2019 04:4...	0

Figure 14. Video records

- 1 Open Videos** Choose the Videos tab and clear any search filter.
- 2 Locate the media** Search by ID, file name, ROV, or time, then select the correct record.
- 3 Play and verify** Choose Play and confirm the content, timestamp, and component context.
- 4 Add or import when authorized** Use Add Video for an approved file, Add Record for metadata-only registration, or Import from DVR db for a supported source.
- 5 Review metadata** Verify ROV, dive number, file path/name, start time, end time, and duration. Refresh after changes.

Figure 15. Remote DVR controller

- 6 Open Remote DV** From DV Connection, choose Open Remote DV. Use the second window only when the approved setup requires it.
- 7 Verify connection settings** Check DVR IP, Local IP, Send Command Port, receive ports, ROV Name, Dive No., online-image folder, and logging path.
- 8 Start listening** Enable Start listening when opened or choose Start Listening. Confirm the status fields update.
- 9 Send the logging path** Choose Send Path after confirming the DVR Logging Path.
- 10 Control recording** Use Start, Stop, and Split only under the approved operational sequence. Observe Last Command and DVR Status after each command.
- 11 Close safely** Stop recording when required, stop listening, save approved defaults if authorized, then Close.

Network caution: Do not change IP addresses or ports during a live survey without coordination with the DVR operator. A wrong endpoint can send commands to the wrong system or prevent logging.

9. Module 8 - Point cloud, PDF, and 3D review

Objective: open visual-reference tools, navigate safely, and associate inspection information without altering controlled source data.

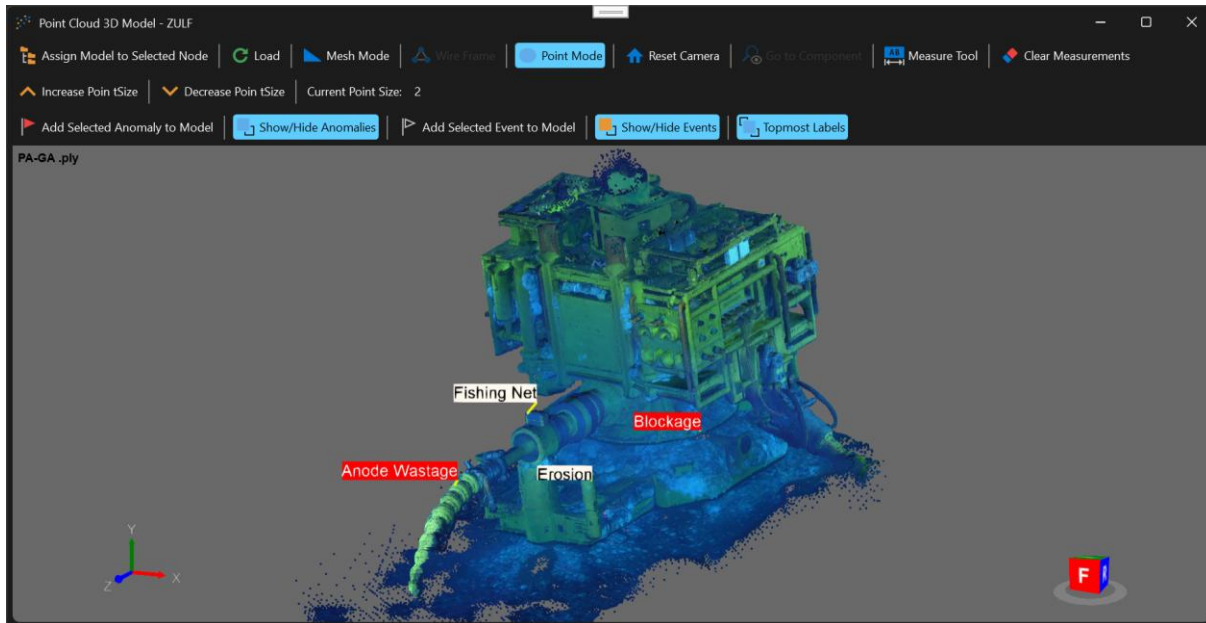


Figure 16. Point Cloud 3D Model

- 1 Open the point-cloud viewer** Select the component and choose Point Cloud Model, then Load the assigned file.
- 2 Choose a render mode** Use Point Mode or Mesh Mode as available. Increase or decrease point size for clarity and use Reset Camera when orientation is lost.
- 3 Measure** Enable Measure Tool, select the required points, record the displayed measurement and model units, then Clear Measurements.
- 4 Show inspection markers** Use Show/Hide Anomalies and Show/Hide Events. Enable Topmost Labels when geometry hides labels.
- 5 Add a selected record** With the correct anomaly/event selected in the main workspace, choose Add Selected Anomaly/Event to Model and place it accurately.

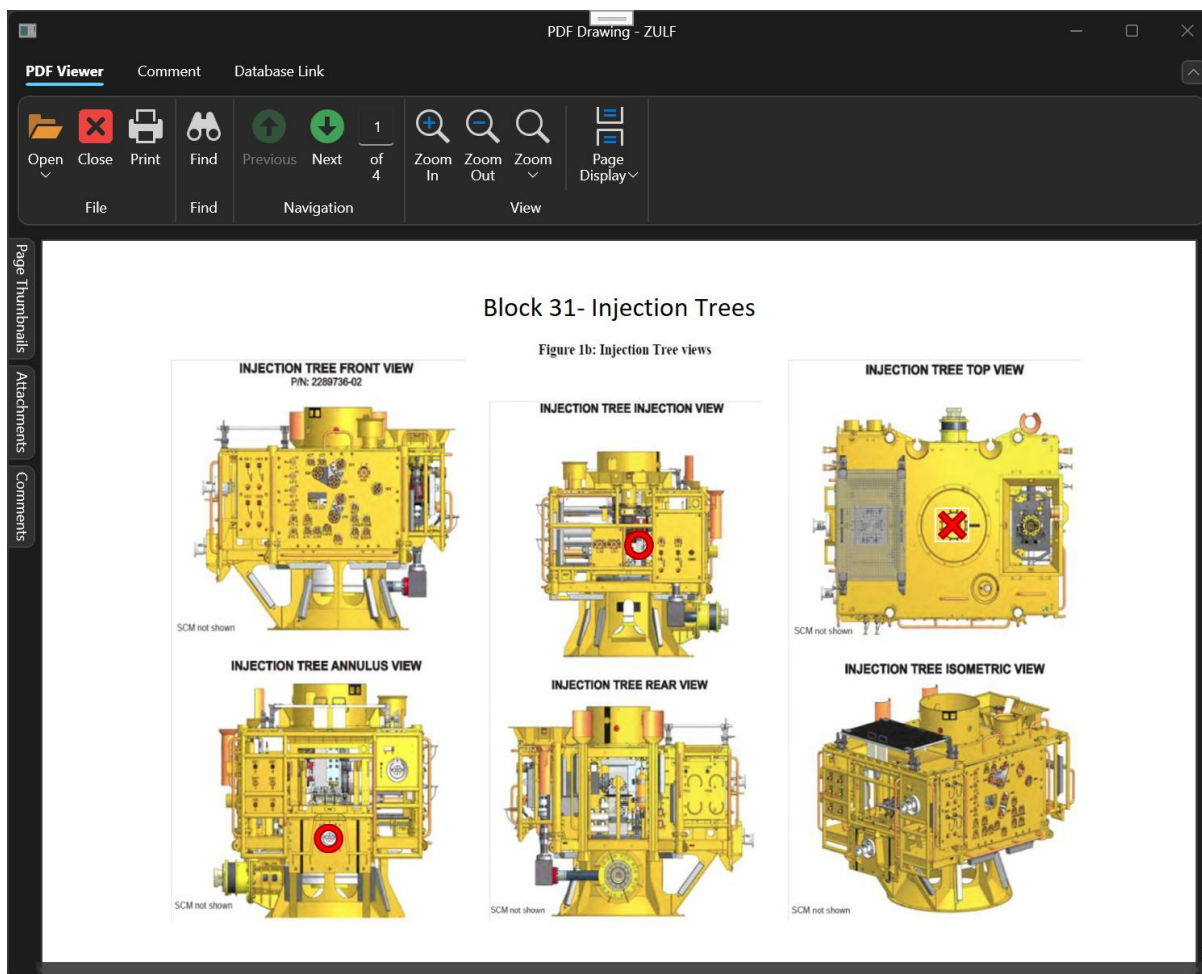


Figure 17. PDF Drawing viewer

- 6 Open the PDF** Select PDF Viewer and choose Open if no drawing is already linked.
- 7 Navigate and review** Use Find, Previous/Next, page number, Zoom, and Page Display. Use the Comment tools only for authorized markup.
- 8 Manage the database link** On Database Link, use Link to associate the PDF with the selected component. Use Un-Link only with authorization; use Save PDF to retain approved markup.

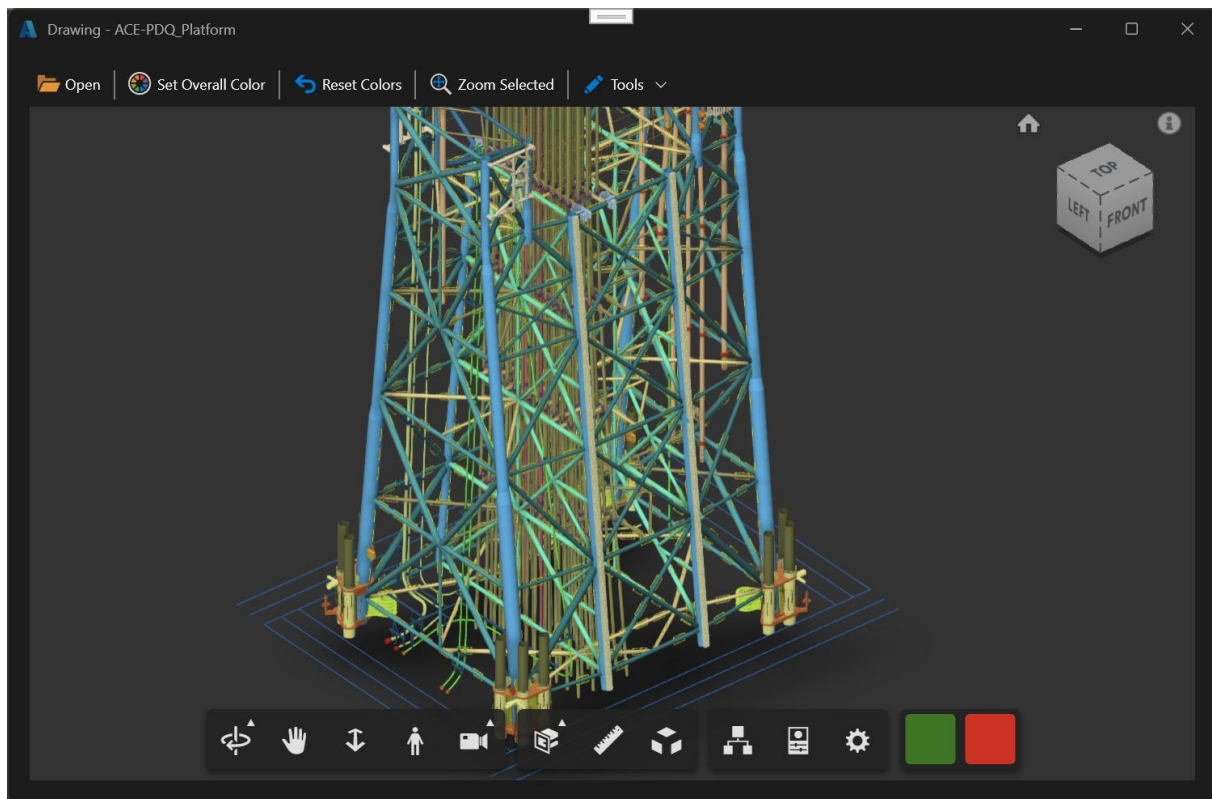


Figure 18. 3D Drawing viewer

9 Open and navigate the 3D drawing Choose 3D Drawing, open the approved model, then orbit, pan, zoom, use the view cube, and Zoom Selected.

10 Use color controls Use Set Overall Color for a temporary presentation view and Reset Colors to restore the model display.

Administrator restriction: Tools > Update Ext.IDs by Known Handles can change many 3D links. It is not a trainee exercise and must be performed only after model/database validation and backup.

PRACTICE: Visual inspection review

Your task: Open one point-cloud model, make and clear a measurement, locate a PDF page, and zoom to a selected 3D object.

Successful when: The trainee navigates all three viewers without changing controlled identifiers or source files.

10. Module 9 - Close and verify the session

Objective: finish work without leaving active tasks, incomplete records, or unverified exports.

- 1 Review active tasks** Confirm every completed task has a valid End time. Record or hand over any task that must remain open.
- 2 Review linked evidence** Open representative task, event, and anomaly images/clips/documents and verify the component and timestamp.
- 3 Refresh key grids** Refresh Tasks, Events, Anomalies, and Videos and confirm saved statuses and record counts.
- 4 Verify status-bar context** Record the final server/database, asset, schedule, and workpack for the handover note.
- 5 Complete approved backup/export** If required, use the approved Backup Database or export procedure. Wait for it to finish and verify the destination file.
- 6 Stop external operations** Confirm no DVR command, video-ID update, copy, clone, import, or export remains active.
- 7 Close the application** Close subsidiary viewer/DVR windows, then close MariSoft-AIS Structures according to local procedure.

Checkpoint: Confirm each item before continuing:

- ☐ No completed task is missing an End time.
- ☐ No unsaved editor window remains open.
- ☐ Required backup/export exists and opens.
- ☐ Incomplete work and errors are included in the handover.

11. Guided end-to-end training scenario

Complete this scenario in the training database without referring to the command-reference manual unless the trainer permits it. The trainer supplies the exact database, component, readings, codes, and media files.

- 1 Connect** to the assigned database and workpack; verify the status bar.
- 2 Navigate** to the assigned component and explain its hierarchy path.
- 3 Create** and start the assigned task record.
- 4 Enter** the supplied structured values and a factual comment.
- 5 Attach** one image and verify it opens.
- 6 End** the task and confirm its time range.
- 7 Create** one event at the supplied video time and attach evidence.
- 8 Convert** the event to an anomaly and complete risk/status fields.
- 9 Add** a follow-up action and set its status.
- 10 Show** the anomaly in the point-cloud model or open its linked drawing/PDF where configured.
- 11 Refresh** all affected grids and complete the session-close checklist.

Pass standard: The complete record chain is traceable from component to task/event/anomaly/action, evidence opens correctly, times are valid, and no unauthorized command was used.

12. Knowledge check

1. Which five context values must be checked before creating a record?

Response: _____

2. What is the difference between a task type and a task record?

Response: _____

3. When should Send Component ID to DVR - On Task Start be used?

Response: _____

4. Why must End Time be later than Start Time?

Response: _____

5. Which fields must be completed after converting an event to an anomaly?

Response: _____

6. What does Can Report control in the Anomaly Images Editor?

Response: _____

7. What should be checked first when a video will not play?

Response: _____

8. Why is Update Ext.IDs by Known Handles restricted?

Response: _____

9. What must be verified after Refresh?

Response: _____

10. Which operations must be allowed to finish before closing the application?

Response: _____

13. Competency assessment and sign-off

No.	Competency	Evidence / acceptance criteria	Result
1	Connect and verify context	Loads the assigned database/workpack and reads back the correct status-bar values.	☐ Pass ☐ Coach
2	Navigate components	Selects the correct component and explains its hierarchy and type.	☐ Pass ☐ Coach
3	Complete a task	Creates/starts/ends a task with valid data, comment, evidence, and times.	☐ Pass ☐ Coach
4	Record an event	Uses approved codes, time, component, and evidence.	☐ Pass ☐ Coach
5	Manage an anomaly	Completes description, classification, risk, status, evidence, and action.	☐ Pass ☐ Coach
6	Use video/DVR safely	Locates and plays media; explains network and recording controls without unsafe changes.	☐ Pass ☐ Coach
7	Use visual tools	Navigates point cloud, PDF, and 3D views and explains restricted commands.	☐ Pass ☐ Coach
8	Close and hand over	Refreshes records, verifies outputs, stops operations, and reports incomplete work.	☐ Pass ☐ Coach

Trainee name: _____ Date: _____

Trainer/assessor: _____ Date: _____

Training database/workpack: _____

Comments / coaching required: _____

Final result: ☐ Competent ☐ Further supervised practice required

14. Quick-reference safety boundaries

- Use Delete and Delete All only with authorization; confirm the selected scope and backup first.
- Do not create SQL databases, change shared templates, copy/clone workpacks, renumber anomalies, update video IDs, or update 3D external IDs during ordinary trainee practice.
- Do not change DVR addresses/ports during a live operation without the DVR operator.
- Do not share database files, media, exports, or screenshots outside approved channels.
- When uncertain, stop, capture the full message and current context, and ask the trainer or system administrator.

Use with the reference manual: For the meaning and prerequisites of every button and menu command, use the MariSoft-AIS Structures Comprehensive End-User Procedure and Command Reference.